Clearroom AI Insights panel: scoring logic design document

Clearroom can compute a robust Deal Risk Score (0–100) by running two parallel algorithmic paths — a deterministic rules-based model for day-one transparency and an ML model that learns from it. The rules path uses a three-dimension weighted composite (completeness 30%, lease risk 45%, financial risk 25%) with circuit breakers for deal-breaking signals, while the ML path uses LightGBM with ordinal-aware loss trained initially on rules-generated pseudo-labels. This document specifies signal definitions, industry thresholds, weight allocations, architecture patterns, and Nordic-specific adaptations across all three panel components: the Deal Risk Score, the "What's Missing" checklist, and the Lease Expiry Concentration chart.

The design draws on FICO composite scoring methodology, Equifax GRESB real estate ESG benchmarking, Envint Basel II/III conservative default principles, INREV institutional reporting standards, Inrev and established CRE risk metrics (WAULT, HHI, ERV gap analysis). No existing PropTech product combines document completeness scoring with lease analytics in a single composite index — Clearroom would be first to market with this approach.

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1. CRE risk signals: definitions, formulas, and thresholds

The institutional CRE due diligence framework revolves around a finite set of quantitative risk signals. Each signal below includes a precise definition, calculation formula, and four-band threshold system calibrated against industry practice from INREV, RICS, and institutional investor norms.

WAULT (weighted average unexpired lease term)

The single most important lease risk metric. **WAULT measures income security duration, weighted by rent or area.** The formula is:

```
WAULT = Σ(Rent_i × RemainingTerm_i) / Σ(Rent_i)
```

Two variants must be computed: **WAULT-to-expiry** (contractual end date, "best case") and **WAULT-to-break** (earlier of break option date or expiry, "worst case"). Grokipedia The gap between these two values is itself a risk signal — a large gap indicates significant break option exposure. INREV defines WAULT-to-break as "weighted average duration of lease contracts based on rent to the shorter of the first tenant break or lease expiry." Inrev

Risk band	WAULT-to-expiry	Signal
Low	>7 years	Strong long-term income security
Medium	3–7 years	Manageable re-leasing exposure
High	1–3 years	Significant near-term vacancy risk
Critical	<1 year	Imminent income instability

Typical WAULT expectations vary by asset class: **logistics expects 5–15 years**, office 3–10 years, anchored retail 5–20 years for anchors with 3–5 years for inline tenants. Capstone Commercial notes that a WALT below 3 years should trigger concern across most asset classes.

Tenant concentration

Measures dependence on a small number of tenants, quantified three ways:

Top tenant as % of rent: `Max(Tenant_Rent_i / Total_Rent) × 100`

Top-3 cumulative rent share: Sum of three largest tenant rent shares.

Herfindahl-Hirschman Index: `HHI = Σ(s_i²)` where s_i is each tenant's fraction of total rent. HHI ranges from near 0 (fully diversified) to 1.0 (single tenant). Open Risk Manual Adapted from antitrust economics, HHI below **0.10** indicates low concentration, **0.10–0.20** moderate, **0.20–0.35** high, and above **0.35** critical.

Metric	Low	Medium	High	Critical
Top tenant % rent	<15%	15–25%	25–40%	>40%
Top 3 tenants	<35%	35–50%	50–70%	>70%
Tenant HHI	<0.10	0.10–0.20	0.20–0.35	>0.35

Anchor tenant dependency deserves special treatment. Anchor vacancy can drop NOI by **30–50%** and trigger cascading co-tenancy abatements in retail assets. NNNTripleNet Single-tenant properties automatically score at the floor of the concentration sub-score.

Lease expiry concentration (expiry cliffs)

Detects dangerous clustering of lease expirations. Computed as the maximum percentage of passing rent expiring in any rolling 12-month window:

```
ExpiryCliff = Max_over_windows(Rent_Expiring_in_Window / Total_Rent × 100)
```

Industry best practice targets no more than **15–20%** of portfolio revenue expiring in any single 12-month period. Theaiconsultingnetwork Anything above 25% is considered high concentration risk and should trigger proactive lease staggering strategies. Theaiconsultingnetwork

Risk band	% rent in any 12-month window
Low	<15%
Medium	15–20%
High	20–30%
Critical	>30%

Break options must be analyzed as potential expiry accelerators. Best practice: model both "contractual expiry" and "worst-case (all breaks exercised)" profiles.

Rent-to-ERV gap

Compares passing rent to estimated rental value: `Gap = (Passing_Rent - ERV) / ERV × 100%`. A negative value signals reversionary upside; a positive value signals over-renting risk (tenants may seek to break or renegotiate at the next review). Forbury

Gap	Classification
>20% over-rented	Critical — significant rent decline ahead
10–20% over-rented	High risk — material reversion exposure
±10%	Low risk — broadly at market
10–20% under-rented	Moderate upside potential
>20% under-rented	Strong reversionary upside

Break option clustering

The concentration of tenant break options in a narrow time window. Formula mirrors expiry cliff analysis but uses break dates: `Break_Concentration = Σ(Rent_of_leases_with_breaks_in_period) / Total_Rent × 100`. Thresholds: <10% low, 10–20% medium, 20–30% high, >30% critical. Breaks coinciding with over-rented leases compound risk materially, as tenants with above-market rents are far more likely to exercise.

Financial signals

Vacancy/Occupancy: Calculated by area (`Vacant_Area / Total_GLA`) or by ERV (INREV-recommended: `ERV_Vacant / (Passing_Rent + ERV_Vacant)`). Property-investment Current US

benchmarks: office **18–20%** vacancy, industrial 6–7%, retail 4–5%. The Kaplan Group For scoring, office vacancy above **15%** and retail/industrial above **10%** triggers concern.

DSCR (Debt Service Coverage Ratio): `NOI / Annual_Debt_Service`. Above **1.50x** is strong, **1.25–1.50x** healthy, below **1.20x** vulnerable, below **1.0x** is default territory. Most lenders require minimum 1.25x.

Service charge recovery: Target **100%** on fully-let properties. Lambertsurv Below 90% on a fully-let property signals lease drafting issues.

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2. Rules-based scoring architecture: the three-dimension composite

The rules-based model uses a weighted arithmetic composite with circuit breaker overrides. This design follows FICO's proven pattern (five weighted categories summing to 100%), Equifax GRESB's management/performance split, Envint and Basel II/III's conservative default philosophy.

Composite formula

```
DealRiskScore = (0.30 × Completeness) + (0.45 × LeaseRisk) + (0.25 × FinancialRisk)
```

Each dimension is normalized to 0–100. The weighted average is chosen over multiplicative aggregation because it allows transparent score decomposition Frontiers (critical for EU AI Act compliance) while circuit breakers prevent dangerous compensability (a perfect lease score should not mask a missing title deed).

Dimension A: completeness score (30% of total)

Sub-component	Internal weight	Scoring logic
Lease doc completeness	40%	`Σ(rent_weight_i × presence_i)` where presence is 1/0 per tenant
Financial doc completeness	20%	Presence of P&L, service charge budgets, rent roll
Technical doc completeness	15%	Environmental reports, building surveys, EPC
Legal doc completeness	15%	Title docs, corporate docs, planning permissions
Amendment chain integrity	10%	`complete_chains / total_chains × 100`, minus 10pts per orphan

Documents are penalized by criticality tier: critical missing documents (title, executed leases, rent roll, Phase I) incur **10-point penalties** per document. Important documents (P&L, PCA, zoning) incur **5-point penalties**. Standard documents (insurance, corporate) incur **2-point penalties**. Quality degradation applies fractional penalties: outdated documents receive 50% of the full missing penalty, incomplete documents 60%, unverified drafts 40%.

The per-category formula is: `Category_Score = Σ(Doc_Quality_Factor) / Expected_Count × 100`, where quality factors are 1.0 (present and current), 0.75 (minor issues), 0.5 (materially outdated), 0.25 (severely deficient), 0.0 (missing).

Dimension B: lease risk score (45% of total)

Sub-component	Internal weight	Scoring function
WAULT/WALT	30%	Piecewise linear: ⩾10yr→100, 7-10→80-100, 5-7→60-80, 3-5→30-60, 1-3→10-30, <1yr→0-10
Expiry concentration	20%	Modified HHI on 12-month rolling windows; >50% in 18mo caps at 20
Tenant concentration	25%	Top tenant >60%→0-15; 40-60%→15-40; 20-40%→40-70; <20%→70-100
Break option risk	15%	`break_gap / WAULT_expiry` ratio; no breaks→100; gap ratio >0.5→20-40
Vacancy indication	10%	⩾95% occupancy→100; linear decline, <70%→0-20

All sub-scores use smooth piecewise linear functions rather than step functions to ensure **score stability** — small input changes produce proportional output changes, a key lesson from FICO model design. Monotonicity is enforced: higher WAULT always produces equal or higher scores.

Dimension C: financial risk score (25% of total)

Sub-component	Internal weight	Logic
Rent level assessment	30%	Over-rented (>110% ERV)→30-50; at market (90-110%)→80-100; under-rented→60-80; no ERV→50 (neutral)
Passing rent concentration	30%	Mirrors tenant HHI but purely financial

Sub-component	Internal weight	Logic
Rent escalation provisions	20%	CPI-linked→90-100; fixed 2-3%→70-90; open market→50-70; none→20-40
Financial doc quality	20%	Presence and recency score

Circuit breakers (critical overrides)

These fire before the weighted average and cap the composite score regardless of other factors:

Condition	Score cap
Missing title/ownership docs	25 (Critical)
Zero leases found when expected	15 (Critical)
WAULT < 6 months	30 (Critical)
Single tenant 100% of rent AND expiry < 12 months	15 (Critical)
>50% expected documents missing	35 (High Risk)
⩾3 orphaned amendments with no base lease	Reduce by minimum 20 points

Implementation: `FinalScore = min(WeightedAverage, LowestCircuitBreakerCap)`.

Risk bands

Band	Score range	Color	Action guidance
Low Risk	75–100	Green 🟢	Proceed with standard DD
Medium Risk	50–74	Amber 🟡	Additional review recommended
High Risk	25–49	Orange 🟠	Escalate to senior team
Critical Risk	0–24	Red 🔴	Deal-breaker signals — urgent review

Bands are deliberately non-uniform: most well-prepared deals cluster in the 50–80 range. Truly "green" deals are uncommon but achievable. The narrow Critical band (0–24) captures only genuinely distressed situations, following FICO's pattern where Bankrate Poor (300–579) spans a wider raw range than Excellent (800–850). Equifax Debt.org

Narrative generation

Template-based NLG produces human-readable risk summaries from rule firings. Each fired rule carries a severity tier, a template string with data bindings, and a computed score impact:

```
CRITICAL: 3 leases expire within 18 months representing 67% of passing rent —
HIGH LEASE EXPIRY CONCENTRATION [Score impact: -28 points]

WARNING: Top tenant (Acme Corp) represents 52% of total rent —
CONCENTRATED TENANT RISK [Score impact: -18 points]

POSITIVE: All 6 leases contain CPI-linked rent escalation —
STRONG RENTAL GROWTH PROTECTION [Score contribution: +6 points]
```

Rules are prioritized by `impact = dimension_weight × sub_weight × score_delta_from_ideal`. The top 5 risk drivers always surface, with any circuit breaker appearing first. Templates support locale-aware formatting for Nordic markets (Swedish/Norwegian/Danish/Finnish) and include data bindings for currency, date, and number formats.

Worked example: "Metro Office Park"

Inputs: 8 expected leases (6 found), all financials present, 1 missing technical report, WAULT-to-expiry 4.2 years, WAULT-to-break 2.8 years, top tenant 38%, top 3 at 72%, 31% of rent expiring within 18 months, all leases CPI-indexed, 92% occupancy, no ERV data.

- **Completeness:** 87/100 (lease gap the main drag)
- **Lease Risk:** 55/100 (moderate WAULT, elevated concentration)
- **Financial Risk:** 69/100 (strong escalation clauses offset by missing ERV data)
- **Composite:** (0.30 × 87) + (0.45 × 55) + (0.25 × 69) = **68 — Amber (Medium Risk)**
- **No circuit breakers triggered.** Top risk drivers: WAULT below 5-year comfort threshold, top-3 tenant concentration at 72%, and 31% expiry concentration.

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3. ML-based scoring path: from rules-bootstrapped labels to production model

The ML path operates in parallel, initially learning from the rules-based model and eventually surpassing it as human-validated labels accumulate. The recommended architecture is **LightGBM with ordinal-aware loss**, chosen because gradient boosting empirically dominates deep learning on tabular data and maintains acceptable accuracy even with datasets below 1,000 examples.

Model selection rationale

The landmark study "Tabular Data: Deep Learning is Not All You Need" (Shwartz-Ziv & Armon, 2021) demonstrated XGBoost outperforms deep models across most tabular datasets. For Clearroom's structured features — document counts, WAULT values, concentration indices — tree-based models are the clear choice. LightGBM specifically offers **native missing value handling** (learns optimal default split directions during training), fast training via GOSS sampling, and exact SHAP computation through TreeSHAP.

For ordinal risk categories (Low/Medium/High/Critical), the model uses **CORAL (Consistent Rank Logits)** loss encoding, which transforms ordinal targets into cumulative binary subtasks with shared weight constraints. This guarantees rank-monotonicity: P(Risk ⩾ Medium) is always ⩾ P(Risk ⩾ High). CORAL is architecture-agnostic and available via `coral-pytorch`.

Isolation Forest runs as an unsupervised companion, detecting unusual dataroom patterns (anomalous document distributions, outlier lease structures) without any labels. This provides immediate value from day one.

Feature engineering specification

Document-level features (6 features): total document count, count per category (lease/amendment/financial/technical/corporate/other), category distribution entropy (`-Σ p_i × log(p_i)`), duplicate ratio, amendment chain completeness ratio, binary flags for absent categories.

Lease portfolio features (8 features): WAULT-to-expiry, WAULT-to-break, tenant count, tenant HHI, max single tenant rent share, lease expiry entropy across time buckets, total break option count, break clustering metric (variance of break dates).

Temporal features (6 features): weighted average time to next break, weighted average time to expiry, % rent expiring within 12/24/36 months, lease vintage distribution statistics (mean, std, skew of start dates).

Completeness features (4 features): overall weighted completeness %, per-category completeness scores, critical document presence flags, field extraction confidence score.

Interaction features (4 features): WAULT × tenant HHI (compounding risk), completeness × lease risk score (uncertainty amplifier), break clustering × near-term expiry %, duplicate ratio × total document count.

Binary "is_missing" indicator features are added for commonly absent fields — the **missingness pattern itself is informative** (missing financial docs signal risk beyond the mere absence of data).

Training data bootstrapping strategy

Phase 1 — Rules as teacher (0 human labels needed): The rules-based model generates soft pseudo-labels (probability distributions over risk categories with temperature scaling T=2–5) for

all deals processed. The ML model trains to minimize KL divergence from these distributions via knowledge distillation. This follows the UKDSL framework demonstrated at production scale.

Phase 2 — Active learning (50–300 human labels): Uncertainty sampling presents deals where ML model entropy is highest to human reviewers. (IntuitionLabs) Diversity sampling ensures labeled examples cover the full feature space. Batch sizes of 20–50 deals per iteration. Active learning typically achieves target accuracy with **30–70% fewer labels** than random sampling. (Label Your Data)

Phase 3 — External data enrichment: MSCI Real Capital Analytics provides 1.8M+ global CRE transactions (CRE Daily) with outcome data (sales, defaults, distress). CMBS loan-level performance data from Morningstar (MSCI) offers rich deal-level outcome labels (performing vs. delinquent vs. default). Public REIT SEC filings provide free property-level performance data for initial development.

Ensemble architecture

```
Rules Score (α=0.8) ──────┐
                          ├── Weighted Ensemble ── Platt Calibration ── Risk
Category + Confidence
ML Score (α=0.2) ─────────┘
```

Alpha starts at **0.80 rules / 0.20 ML** and shifts toward ML as it proves itself on held-out human-labeled validation data. Disagreement flagging triggers human review when models diverge by more than one risk level — these disagreements are the most informative candidates for active learning.

Calibration uses Platt scaling (logistic regression on classifier outputs) (Wikipedia) — only 2 parameters to learn, robust with <1,000 calibration examples. (Wttech) Evaluated via Expected Calibration Error (ECE) and reliability diagrams. (Train in Data)

Performance targets and evaluation

Metric	v1 target	Rationale
AUC-ROC	0.70–0.80	Achievable with <1,000 tabular examples
Macro F1	0.55–0.65	4-class ordinal with imbalanced classes
Ordinal MAE	<0.5 risk levels	Predictions within half a category on average
ECE (calibration)	<0.10	After Platt scaling
Precision@10% (Critical)	>0.70	Most operationally important metric

Minimum viable dataset: 200–300 labeled deals for meaningful XGBoost with ~15 features (expect AUC ~0.65–0.72). **500–700 labeled deals** is the sweet spot for v1. With rules-generated pseudo-labels, ML training can begin with **zero human labels** and 500+ rule-labeled deals.

Evaluation uses **nested 5-fold cross-validation** (inner fold for hyperparameter tuning, outer for unbiased performance estimation) (scikit-learn) with stratified splits preserving class distribution. (Analytics Vidhya) McNemar's test confirms statistical significance of performance differences between rules and ML paths.

Production deployment

Shadow mode first: ML scores are computed and logged alongside rules scores but only rules scores surface to users. After validation, gradual rollout shows ML scores to 10% of users. Model drift is monitored via Population Stability Index (PSI > 0.1 triggers alert) on feature distributions weekly. Retraining occurs quarterly minimum or upon drift detection, using LightGBM's incremental training capability.

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4. Lease expiry concentration chart: measurement and visualization

Time bucket specification

The primary view uses **annual buckets** for the first 5 years plus an aggregated 5+ year bucket, consistent with institutional investor reporting conventions:

- Year 1 (0–12 months)
- Year 2 (12–24 months)
- Year 3 (24–36 months)
- Year 4 (36–48 months)
- Year 5 (48–60 months)
- 5+ years (aggregated)

A secondary near-term view provides quarterly granularity for Years 1–2 and annual thereafter, supporting operational decision-making. For multifamily contexts, monthly bucketing matters because seasonal leasing patterns affect no single month should exceed **15%** of total expirations.

Weighting: passing rent as primary, area as secondary

Passing rent is the industry standard for institutional expiry analysis because it directly reflects cash flow at risk. (A.CRE) Area-based weighting serves as a secondary lens for operational/leasing teams. ERV-based weighting (INREV-recommended for reversionary analysis) can be offered as an optional third view when ERV data is available. (Property-investment) Unit count weighting applies only to multifamily.

Dual-profile visualization

Every expiry chart should present **two profiles simultaneously:**

1. **Contractual expiry profile** (solid bars): uses lease end dates
2. **Break-adjusted profile** (hatched overlay): uses earlier of break date or expiry per lease

This dual view, consistent with INREV WAULT-to-break methodology, reveals the gap between contractual and worst-case scenarios. The delta between profiles is itself a key risk indicator.

Concentration risk flagging

A **horizontal threshold line at 20%** of total rent marks the concentration danger zone. Bars exceeding this threshold are highlighted with a red border. The HHI of the expiry distribution is computed: `HHI = Σ(bucket_share²)`. For a perfectly staggered 5-bucket profile (20% each), HHI = 0.20. Values above **0.25** trigger a concentration risk flag. A secondary metric — max single-bucket percentage — is the simplest and most widely used in practice.

Color coding

Time horizon	Color	Risk signal
0–12 months	Red/Orange	Immediate rollover risk
12–24 months	Amber/Yellow	Near-term risk
24–36 months	Yellow/Light green	Moderate horizon
3–5 years	Green	Medium-term stability
5+ years	Dark green/Blue	Long-term secured income

Visualization format: **stacked bar chart** (industry standard used by Argus, CompStak, Adventures in CRE) with bars stacked by tenant to show which tenants contribute most to each bucket. A cumulative percentage line on the secondary axis shows the income decay curve.

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5. "What's Missing" checklist: dynamic generation by deal type

Canonical document taxonomy

The master checklist spans 15 categories organized into four criticality tiers, synthesized from Thompson Coburn, Adventures in CRE's 72-task acquisition checklist, A.CRE RICS TDD standards, and European institutional practice:

Tier 1 — Critical (10× weight): Title/ownership documents (deeds, land registry, encumbrances), executed leases with all amendments, certified rent roll with arrears schedule, tenant estoppels.

Tier 2 — Important (7× weight): Financial statements (trailing 3-year P&L, service charge reconciliations, capex history), environmental assessments (Phase I ESA or Nordic equivalent), technical/building condition reports (PCA, M&E reports, floor plans), planning/zoning compliance documentation.

Tier 3 — Standard (4× weight): Insurance certificates and claims history, corporate/entity formation documents, property tax bills and assessments, property management and service contracts, financing/loan documents (if assuming debt).

Tier 4 — Supporting (2× weight): Tenant correspondence (rent reviews, break notices, dilapidations), prior valuations and market analysis, legal/litigation schedules.

Dynamic expected counts

Document expectations are computed from deal metadata rather than fixed lists:

```
Expected_Leases = Number_of_Tenants (derived from rent roll)
Expected_Amendments = Known_amendment_chains × expected_amendments_per_chain
Expected_Environmental = Number_of_Buildings × 1 (Phase I per building)
Expected_Tax_Bills = Number_of_Parcels × 3 (years)
```

This dynamic approach — which no existing PropTech tool currently offers — ensures the checklist adapts to each deal's complexity.

Asset class variations

Office adds: fit-out agreements, BMS documentation, BREEAM/LEED/Miljöbyggnad certification, lift/elevator inspection certificates, telecom infrastructure docs.

Retail adds: turnover rent provisions, co-tenancy/anchor clauses, CAM reconciliations, signage agreements, operating hours restrictions, food/beverage licensing.

Logistics/Industrial adds: environmental compliance documentation (extensive), hazardous materials permits, floor load capacity certifications, clear height documentation, contamination history, fire suppression system certificates.

Mixed-use: union of all applicable asset class categories, organized by use component.

Completeness calculation

```
Weighted_Completeness = Σ(Category_Score_i × Category_Weight_i) /
Σ(Category_Weight_i) × 100
```

```
Where: Category_Score_i = Σ(Doc_Quality_Factor_j) / Expected_Count_i
```

Quality factors: present and current = **1.0**, minor issues = 0.75, materially outdated (>12 months) = **0.5**, severely deficient = 0.25, missing = 0.0.

Completeness bands: **90–100%** Green (substantially complete), **70–89%** Amber (material gaps), **50–69%** Yellow (significant gaps, DD unreliable), **below 50%** Red (major information requests required).

UX recommendation: hybrid priority + category view

The primary view is a **priority-ranked missing items list** sorted by `criticality_tier × category_weight × score_impact`, with Critical items (red) at top and action buttons to generate information request letters. A secondary drill-down view groups by category with per-category progress rings and traffic light indicators following the **RICS Red/Amber/Green standard** Property Inspect — the most widely recognized risk communication system in European CRE. Contextual intelligence annotations explain findings in plain language: *"Based on 10 tenants in rent roll, we expect 10 leases — only 7 found"* or *"Phase I ESA is 3 years old — may need updating for lender requirements."*

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6. Nordic and European adaptations

Swedish lease parsing: Fastighetsägarna Form 12B

The Form 12B SEC.gov (revised as 12B.2 circa 2011) is the industry-standard Swedish commercial lease template. Clearroom's extraction engine must parse these specific fields:

- **Hyrestid** (lease term): fixed-term, typically 3–10 years; max 25 years within detaljplan areas Dlapiperrealworld Baker McKenzie
- **Hyra** (base rent): SEK/year, quarterly in advance standard
- **Uppsägningstid** (notice period): typically 9–12 months Baker McKenzie
- **Indexklausul** (CPI clause): October KPI as reference month, standard for leases ⩾3 years Dlapiperrealworld
- **Fastighetsskatteklausul** (property tax pass-through via Form 7B supplement)
- **Säkerhet** (security deposit/bank guarantee): typically 3–6 months' rent
- **Särskilda villkor** (special provisions overriding pre-printed terms) SEC.gov

Indirect besittningsskydd (indirect possession protection)

This uniquely Swedish mechanism has **direct scoring implications**. After 9+ months of tenancy, tenants gain a statutory right to compensation (minimum 1 year's rent) if the landlord terminates without valid cause Lexology — but critically, the tenant **cannot force lease extension**

Baker McKenzie (unlike UK security of tenure). Valid termination grounds that avoid compensation include tenant breach, demolition/major renovation, and permitted change of use. Dlapiperrealworld

WAULT calculation impact: Swedish auto-renewal mechanics (leases extend for the same term if neither party gives timely notice) Prima Law combined with the compensation barrier mean **actual lease duration often exceeds contractual terms**. The scoring model should account for this by applying a Swedish "effective WAULT multiplier" that extends expected duration by the probability of non-termination.

Tomträtt (site leasehold) risk

Unique to Swedish property: an indefinite right to use publicly-owned land Lexology Dlapiperrealworld with ground rent (tomträttsavgäld) renegotiated every 10+ years. The next renegotiation date and exposure to ground rent increases must be captured as a distinct risk signal. Properties held under tomträtt rather than freehold may face investor mandate restrictions.

CPI indexation analysis

Nearly all Nordic commercial rents are **CPI-indexed annually** CE Sweden — a fundamental structural difference from UK (upward-only market reviews) and US (stepped increases or capped CPI). The 2022–2024 high-inflation period produced ~10% annual rent jumps in Sweden. Clearroom should flag leases with non-standard CPI provisions: below-market floors, caps that limit escalation, or absence of indexation entirely. Pure CPI pass-through is the benchmark.

Nordic DD document checklist additions

Beyond the standard international checklist, Nordic transactions require:

- **Fastighetsregisterutdrag** (Swedish land registry extract from Lantmäteriet)
- **Energideklaration** (Swedish EPC, A–G rated, valid 10 years, mandatory for leased buildings) Baker McKenzie
- **Detaljplan** (detailed development plan/zoning documents)
- **Brandskyddsdokumentation** (systematic fire safety documentation)
- **Miljöundersökning** (Phase I equivalent; contaminated land check against Länsstyrelsen register)
- **Pantbrev register** (mortgage certificates)
- **Grunnbok** (Norwegian land register from Kartverket)
- **Meglerstandard** lease forms (Norwegian broker standard, 6th edition 2019) DLA Piper
- **Energimærkning** (Danish energy label, mandatory for transfers/leases) Deloitte

European regulatory requirements

EU Taxonomy/SFDR compliance is now expected in datarooms for Article 8/9 funds. Required

documentation includes: EU Taxonomy alignment assessment per asset, energy consumption data (kWh/m²/year), carbon emissions data ($kgCO_2/m^2$/year), **CRREM pathway alignment** analysis (country-specific, property-type-specific 1.5°C decarbonization pathways with "stranding date" computation), Crrem and building certifications (BREEAM, Miljöbyggnad in Sweden, DGNB in Denmark).

EPC regulatory risk: Sweden's Boverket set a preliminary 2033 MEPS (Minimum Energy Performance Standards) threshold at **174 kWh/m²/year** for commercial buildings. Buildings rated F or G face stranding risk and should be flagged. The scoring model should include an ESG/climate risk sub-component for European deals.

AML/KYC documentation is mandatory across all Nordic jurisdictions under EU Anti-Money Laundering Directives. Swedish SPV transactions (aktiebolag) require full beneficial ownership chain verification.

Transaction structure: share deal vs. asset deal

Most Nordic CRE transacts via **share deals** (selling the SPV) to avoid stamp duty (4.25% for legal entities in Sweden vs. 0% for share transfers). This means DD covers both company-level and property-level diligence. Clearroom's checklist should detect transaction structure and add corporate/entity document requirements accordingly.

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7. Auditability, explainability, and regulatory compliance

Score decomposition

Every score includes a **waterfall decomposition** starting from 100 and showing each deduction with its cause, the three dimension scores with weighted contributions, a full rule audit trail with timestamps and input data, and data provenance linking each data point to its source document. This satisfies EU AI Act Article 13 (transparency) and Article 14 (human oversight) requirements.

EU AI Act positioning

CRE deal-level risk scoring (scoring properties, not individuals) likely does not fall under the "high-risk" classification of Annex III Section 5(b), which targets creditworthiness assessment of natural persons. However, designing the system with high-risk compliance from the outset provides maximum regulatory safety and customer confidence. The rules-based path is inherently a "white box" — the most transparent possible design. The ML path uses TreeSHAP for post-hoc explanations, which satisfies explainability requirements when combined with comprehensive technical documentation.

Version control

Scoring rules use semantic versioning (Major.Minor.Patch). Every deal scored retains the exact rule version used, creating an **immutable audit trail**. A/B testing capability allows two rule

versions to run in parallel before promoting changes.

Conclusion: implementation roadmap and novel insights

This design document specifies a scoring system with several properties not found in existing PropTech products. The **dual-path architecture** (rules + ML in parallel) solves the cold-start problem: rules deliver value on day one while generating training signal for the ML model. The **circuit breaker pattern** borrowed from Basel III prevents dangerous score compensability. The **dynamic expected document count** based on deal metadata (tenant count, building count, jurisdiction) surpasses the static checklists used by current DD tools.

Three implementation insights emerged from this research. First, the gap between WAULT-to-break and WAULT-to-expiry is an underexploited risk signal — the **break delta ratio** (`gap / WAULT_expiry`) deserves its own sub-score and narrative template. Second, Swedish indirect besittningsskydd creates an asymmetry where effective lease duration exceeds contractual terms; modeling this requires a jurisdiction-specific WAULT adjustment that no current analytics platform provides. Third, the ML model's highest-value early contribution is likely **anomaly detection** via Isolation Forest rather than supervised risk scoring — flagging unusual dataroom patterns provides immediate insight without any labeled data.

The recommended build sequence is: rules-based scoring engine with narrative generation (weeks 1–6), completeness checklist with dynamic expected counts (weeks 3–8), lease expiry visualization (weeks 4–6), Isolation Forest anomaly detection (weeks 5–8), LightGBM training on pseudo-labels with SHAP integration (weeks 7–12), shadow mode ensemble deployment (weeks 9–12), and active learning label collection as an ongoing process. Alpha weighting should start at 80/20 rules/ML and shift based on human-validated performance comparisons.